

Mahmoud Shadeed

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EDUCATION

UNIVERSITY OF MICHIGAN

Bachelor of Science, Major In Computer Science

Ann Arbor, MI

2021- 2025

EXPERIENCE

BLUE CROSS BLUE SHIELD OF MICHIGAN

Detroit, MI

Data Engineer Intern

May 2024-August 2024

- Developed and implemented data quality checks and analyzers using Python 3 and PySpark, enabling reusable, fast, and effective cross-team business decisions.
- Created a unified Python and PySpark-based database connection library, simplifying access to MongoDB, Neo4j, and PostgreSQL, and streamlining ETL workflows and data integration.
- Implemented text analysis Spark Databricks notebooks using data from sensitive medical documents for future use in NLP models, including Named Entity Recognition (NER) and text classification, to advance data-driven insights.
- Languages Used: Python, PySpark, SQL, Scala, Shell

PERSONAL PROJECTS

Data Structures and Algorithms | C++

- Gained a strong foundation in Object Oriented Programming in C++.
- Implemented a priority queue system utilizing inheritance and polymorphism, including sorted, binary heap and pairing heap implementations.

Machine Learning Models | C++

- Gained experience with recursion, binary trees, templates, comparators, and the map data structure.
- Developed the Bag of Words Machine Learning Model from scratch in C++, enabling developers to use the model in search engines.

Collaborative Filtering Pipeline for Movie Ratings | PySpark, Databricks

- Transformed a movie ratings CSV into a PySpark DataFrame and trained a collaborative filtering model using the ALS algorithm on Databricks.
- Tuned model parameters, evaluated accuracy using a regression evaluator, and built a scalable pipeline to generate predictive movie rating recommendations.

Scala Twitter Streaming API | Python, Scala

- Developed skills in Scala and became proficient in utilizing the Twitter Streaming API for developing real-time data processing solutions.
- Implemented a local Spark streaming service for large scale data of Tweets from Twitter API for analysis of tracking average tweet lengths and most popular Hashtags.

ADDITIONAL

SKILLS

- **Languages:** C++, Java, SQL, PL/SQL, Unix Shell, JavaScript, Excel Macro, Scala, Spark, Typescript
- **Frameworks and Tools:** PostgreSQL, Redshift, MongoDB, DynamoDB, Git, SQL Server Integration Services, Informatica, Oracle SQL Loader
- **Certifications:** SPARK Programming with Scala and PySpark

AWARDS AND LEADERSHIPS

- **2021 - Present:** Phi Sigma National Honor Society at the University Of Michigan-Ann Arbor
- **2020 - 2021:** PHI THETA KAPPA Achievement at Washtenaw Technical Middle College